

Nevò MIRZAI HAMADANI

 nevomirzai.com |  nevo.mirzai@gmail.com |  linkedin.com/in/nevò-mirzai

EDUCATION

EPFL | Master in Robotics (2nd year)

Sep. 2024 – July 2026

- Minor in Data Science: Reinf. Learn.: 6/6, Deep Learn.: 5.75/6, Image analysis: 5.75/6
- Relevant courses: Robotic Manip.: 5.5/6, Aerial Robotics: 5.5/6, Legged Robots: 5.25/6

EPFL | Bachelor in Mechanical Engineering

Sep. 2021 – July 2024

- 1 year exchange at the Chinese University of Hong Kong (CUHK)
- Relevant courses: Linear Algebra: 5.75/6, Analysis: 5.25/6, Algorithms: 5.75/6

TECHNICAL SKILLS

 **Programming** Python, C++, C, MATLAB, C#

 **Simulation** IsaacSim, IsaacLab, OpenAI gym, Unity, Genesis, Webots, MuJoCo, Gazebo

 **Computer Vision** SLAM (Visual/LiDAR), Image Analysis, OpenCV

 **Machine Learning** Reinforcement Learning, Deep Learning, Pytorch

 **Technical** ROS2, CUDA, Git, IK, Optimization, Hybrid Control, MPC, LQR

 **Laboratory** Soldering, electrical circuits, CAD, 3D Printing

EXPERIENCE

Student Mentor

Sep. 2022 – Jan. 2026

- Supporting freshmen during their academic year with advice and practical help.

Teaching Assistant

Sep. 2022 – June 2024

- Mechatronics, Electronics, Linear Algebra, Calculus, Physics I, Thermodynamics, Algorithms, Programming, Mechanical Behaviour of Materials, Numerical Analysis.
- Supporting students during exercise sessions.

EPFL Rocket Team Engineer

Sep. 2021 – June 2024

Vertical landing rocket research project (Icarus) - Simulink

- Simulated vertical landing of our rocket in Simulink. Currently testing for Sim2Real.

RESEARCH PROJECTS

Imitation Learning with Robotic Arm - LeRobot, Teleoperation,

- Assembled a leader-follower robotic arm system, controlled on Ubuntu.
- Implemented an ACT (Action Chunking with Transformers) policy.
- Designed a vision-based control pipeline using ResNet-18 encoders for wrist-mounted camera input and Temporal Ensembling for trajectory generation.

Hand Gesture Drone Swarms Control - Sim2Real, ROS, C#, Python

- Conceived an intuitive system for controlling a swarm of drones using hand gestures.
- Created and controlled a simulated swarm in Unity, and applied ROS to control a real swarm (Digital-Twin).

Soft Robotic Arm: Sim2Real Digital Twin & RL - Isaac Sim, Sim2Real

- Deployed a Digital Twin of a soft robotic arm using NVIDIA Isaac Sim.
- Approximated soft dynamics with rigid-body physics and calibrated stiffness/damping.
- Modeled an IMU sensor behaviour for accurate simulation data.

PROJECTS

Reinforcement Learning for Quadruped Robot - RL, OpenAI Gym, Isaac Gym

- Trained a SAC Reinforcement Learning policy to achieve quadruped robot locomotion, including high/low velocity gait control and stair climbing.
- Investigated and integrated movement algorithms (Central Pattern Generators (CPG)).

Robotic Arm Simulation for Scaffolding Applications in Hong Kong - Isaac Sim

- Simulated inverse kinematics for a robotic arm to assist in scaffolding construction.

Inverse RL research - Reinforcement Learning

- Conducted research into the limitations and possible solutions of current IRL methods.

Autonomous drone racing - Path planning, Hardware

- Programmed a drone to race a given trajectory autonomously, first in simulation and then deployed on a real Crazyflie (Sim2Real). Path planning, Computer Vision.

Billiard Game Analysis - Computer Vision, C, Simulink

- Assessed billiard game dynamics using MATLAB, C, and LabVIEW to detect the winner.

Sensor Noise Characterization & Synchronous Detection - Hardware

- Analyzed sensor noise spectrally, implemented ambient light rejection.

Autonomous Vehicle Navigation System Using MPC - Control, Simulink

- Devised a simplified autonomous vehicle navigation system in simulation (MPC).

Eye Tracking Headlamp - Robotic Manipulation, Computer Vision

- Prototyped a Raspberry Pi-based headlamp using dual IR/optical cameras for real-time gaze tracking.
- Engineered a closed-loop system to automatically align lighting with the user's eyes.

Data Creation Box - Object Recognition, System Integration

- Developed an automated physical environment to generate diversified organic datasets. Then fine-tuned object recognition models (YOLO). Deployed on a robotic arm system, using a force-torque sensor and a camera.

AI model training for Hate Speech prediction - GPT2/BERT Fine-tuning

- Optimized GPT-2 and BERT to predict the probability that hate speech will be used in a phrase given the context, then compared the results.

Clickbait data analysis: Is clickbait recognizable from data? - Pandas, Pytorch

- Analyzed metadata from YouTube and developed indicators of clickbait videos.

Chocolate recognition - Computer Vision

- Leveraged Computer Vision to recognize and count the number of different chocolates in multiple images with varying backgrounds.

Obstacles Dodging and Navigation for a Robot - State Estimation, CV, Path plan.

- Implemented vision (OpenCV), Kalman filter, and path planning for a Thymio robot.

Quadcopter Design and Stabilization - Robotic Manipulation, Control, Design

- Programmed PID stabilization from scratch using an MPU and an Arduino.

AWARDS - INTERESTS - LANGUAGES

 Guinness World Record (2017): Most drones built and flown simultaneously (U18)

 Judo (19 years experience, pre-professional level)

 English C1, French C1, Italian C2, German B1

Prof Mirko Kovač

Director, Laboratory of Sustainability Robotics
Empa - Swiss Federal Laboratory for Materials Science
and Technology
EPFL - École Polytechnique Fédérale de Lausanne

Tel.: CH +41 (0)58 765 1122
E-mail: mirko.kovac@empa.ch



Zurich, 18.11.2025

Reference Letter for Mr Nevo Mirzai Hamadani

To Whom It May Concern,

I am pleased to provide this reference for Mr Nevo Mirzai Hamadani, who completed his semester project under my supervision at the Laboratory of Sustainability Robotics at Empa and EPFL. The project focused on developing a simulation framework for Aerial Additive Manufacturing, involving the evaluation of physics engines and material models. This required a combination of analytical thinking, programming skills, and a structured approach to problem-solving.

Nevo performed well throughout the project, demonstrating a solid understanding of the core concepts. He worked independently, managed his tasks responsibly, and consistently demonstrated a willingness to learn new tools and methods. His final project showed good overall performance, and he had steady engagement with the work.

Nevo communicates clearly, is receptive to feedback, and can adapt his approach when needed. He handled challenges with a practical mindset and maintained good working discipline. While further experience will help him deepen his technical expertise, he has shown the foundations of a capable engineer with the potential to grow in academically or professionally demanding environments.

I am confident that Nevo will continue to develop positively in his future endeavours. Please feel free to contact me if further information is required.

Yours sincerely,

Yusuf Furkan Kaya

(On behalf of the Laboratory of Sustainability Robotics)

Lausanne, August 2025

Reference letter for Nevò Mirzai Hamadani

To Whom it May Concern

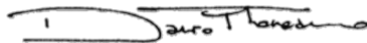
It is a pleasure to write this letter of recommendation for Nevò Mirzai Hamadani. This letter was written in collaboration with Benjamin Jarvis, a PhD student in my lab who advised and worked closely with Nevò. I have reviewed, edited, and approved its contents.

Nevò completed a semester project in my lab during the Spring of 2025. His project focused on developing a method for controlling a swarm of quadrotors using hand pose information gathered from a virtual reality headset.

Nevò implemented a system to capture hand pose data from a Meta Quest 3S headset into a Unity simulation. He devised a method to map the hand pose into a 3D ellipsoid that reflected the user's hand movements. This ellipsoid was then used to adjust the parameters of a swarming algorithm, enabling intuitive control over the swarm's shape and velocity. The approach was validated on a real swarm of Crazyflie drones running on a ROS2 architecture. To achieve this, Nevò established a UDP connection to transmit swarm parameters to the server controlling the drones.

Nevò was a capable and committed student who successfully implemented the proposed approach and explored alternative methods for swarm control. He was also successful in demonstrating the project results with a swarm of real drones.

Yours sincerely



Prof. Dario Floreano